

Pooja Kiran Bharadwaj

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Education

Arizona State University

Master of Science in Information Technology

Tempe, AZ

Aug. 2024 - May 2026

- Coursework: Machine Learning Security, Cloud Security Architecture, Information Assurance, Network Defense, Secure Systems

M. S. Ramaiah University of Applied Sciences

Bachelor of Technology in Computer Science and Engineering

Bengaluru, India

Aug. 2019 - Aug. 2023

Experience

AI Security Researcher & Engineer

Aug. 2024 - Present

Independent Security Research (Concurrent with M.S. program)

Tempe, AZ

- Engineered an inline MCP security gateway with a 5-layer decision pipeline (server trust, tool-call policy, process-spawn evaluation, semantic intent, network egress) and 55 prompt-injection detection rules, implementing JSON-RPC parsing, Unicode/homoglyph normalization, and real-time policy enforcement in Python/FastAPI, achieving sub-millisecond average inspection latency across the full pipeline
- Developed a static IAM policy analyzer for AI agent roles on AWS with 24 deterministic rules covering wildcard permissions, privilege escalation, audit-trail tampering, and credential-harvest chains with SARIF output, enabling automated least-privilege enforcement as a CI merge gate
- Built a model supply-chain scanner with a taint engine and symbolic resolver covering 8 file format families (pickle, safetensors, GGUF, ONNX, H5/Keras, config, Python) by implementing deep opcode analysis for pickle Protocol 0-5, achieving 12/12 detection on documented real-world attacks (CVE-2024-5480, CVE-2024-25664) with 0% false positive rate
- Created adversarial prompt-generation and model-extraction attack harnesses (FGSM, PGD, C&W) mapped to MITRE ATLAS, producing reproducible security assessments with structured SARIF evidence output for guardrail validation and input/output filtering decisions

Business Compliance Lead & Market/Cost Analyst

Aug. 2025 - Dec. 2025

AEROSEC - ASU Technology Innovation Lab (Honeywell Aerospace)

Tempe, AZ

- Led regulatory compliance analysis for Honeywell Aerospace's security product by mapping requirements and conducting competitive cost modeling across aerospace frameworks, delivering a compliance framework supporting the go-to-market strategy

Graduate Teaching Assistant / IT Grader

Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Tempe, AZ

- Evaluated technical assignments for 100+ students per semester across cloud security architectures, network defense configurations, and IAM access control implementations

Projects

MCP Agent Security Gateway | Python, FastAPI, Docker, Kubernetes, GitHub Actions

- Architected an inline stdio proxy intercepting MCP tools/call JSON-RPC requests through a 5-layer pipeline with allow/block/redact/quarantine decisions, preventing unauthorized tool execution before requests reach downstream servers
- Implemented 55 prompt-injection detection rules with Unicode and homoglyph normalization, PII detection for 9 data types (email, SSN, credit card, phone, IP, DOB, passport, AWS keys, API keys), and exfiltration-pattern matching
- Engineered production-grade operational controls: SHA-256 hash-chained audit logging, circuit breakers, rate limiting, shadow mode, and write-ahead logging with Kubernetes deployment manifests
- Validated with 698 automated tests at 90% coverage across Python 3.10/3.11/3.12, integrating Bandit, CodeQL, Trivy, Gype, and pip-audit into the CI pipeline

AWS Agent Identity Guard | Python, AWS IAM, Terraform, SARIF

- Developed 24 deterministic IAM rules targeting AI agent roles by detecting wildcard service prefixes, iam:PassRole without conditions, privilege-management actions, audit-trail tampering, and credential-harvest plus lateral-movement chains for Bedrock, SageMaker, and Lambda workloads
- Implemented credential-harvest chain detection analyzing combined permission patterns across sts:AssumeRole, cross-account trust without ExternalId, and network egress modifications
- Built a CI merge gate outputting findings in Text, JSON, and SARIF formats with exit code 1 on critical/high severity, with Terraform infrastructure examples for deployment

HF Model Provenance Scanner | Python, Hugging Face API, Docker, SARIF

- Engineered a taint engine and symbolic resolver for pickle opcode analysis (Protocol 0-5) implementing REDUCE/GLOBAL/STACK_GLOBAL walks with gadget-chain matching, detecting memoized-global exec and importlib bypass techniques
- Built temporal baseline diffing for rug-pull detection with SafeTensors header inspection, GGUF structural parsing, typosquat detection via Levenshtein distance, and multi-layer obfuscation decode (base64, chr(), string concat)
- Reduced scan bandwidth by 99.9% using HTTP Range requests to fetch only first 512KB of opcodes and metadata headers, scanning GPT-2 in 0.5MB instead of downloading 500MB of tensor weights
- Added runtime inference monitor (intercepts torch.load before execution), cryptographic provenance ledger (hash-chained Ed25519-signed event log), and model quality evaluator (bias, drift, accuracy). 350 tests passing, scanned top 100 HuggingFace models

LLM Red-Team Framework | Python

- Built adversarial prompt-generation and detector-evaluation framework implementing multi-category attack generation following OWASP LLM Top 10, producing per-category detection-rate reports with structured JSON evidence

Publication

A Personalized E-Learning System Using Reinforcement Learning Through Satellite
IEEE INDICON 2023 | IEEE Xplore: 10440852

Technical Skills

AI/ML Security: Agentic AI Security, LLM/RAG Security, Prompt Injection & Jailbreak Testing, MCP/Tool Security, Model Supply-Chain Security, Adversarial ML (FGSM, PGD, C&W), Data Poisoning, Membership Inference

Standards: MITRE ATT&CK/ATLAS, OWASP Top 10 for LLM Applications, NIST AI RMF, CycloneDX SBOM

Cloud & IAM: AWS IAM, Least-Privilege Enforcement, Trust Relationships, Agent Role Security, Terraform

Languages & Tools: Python, FastAPI, PyTorch, scikit-learn, Docker, Kubernetes, GitHub Actions, CI/CD, SARIF

Security Eng.: Threat Modeling, Red Teaming, Guardrails Design, Input/Output Filtering, Access Controls, Audit Logging

Certifications & Recognition

AWS Academy Graduate: Cloud Security Foundations | Technology Innovation Lab: Honeywell Aerospace & ASU | KSCST: Research Grant